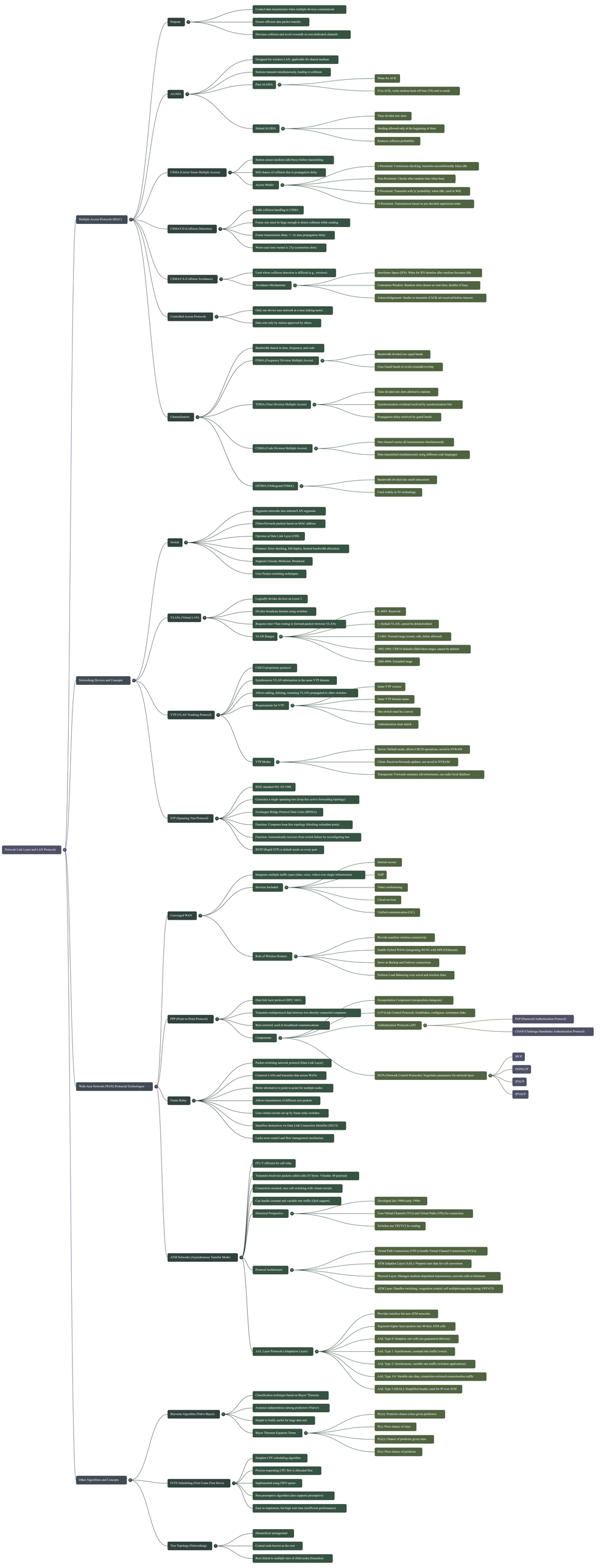
Network Link Layer and LAN Protocols
Multiple Access Protocols (MAC)
Purpose
Control data transmission when multiple devices communicate
Ensure efficient data packet transfer
Decrease collision and avoid crosstalk on non-dedicated channels
ALOHA
Designed for wireless LAN, applicable for shared medium
Stations transmit simultaneously, leading to collision
Pure ALOHA
Waits for ACK
If no ACK, waits random back-off time (Tb) and re-sends
Slotted ALOHA
Time divided into slots
Sending allowed only at the beginning of slots
Reduces collision probability
CSMA (Carrier Sense Multiple Access)
Station senses medium (idle/busy) before transmitting
Still chance of collision due to propagation delay
Access Modes
1-Persistent: Continuous checking, transmits unconditionally when idle
Non-Persistent: Checks after random time when busy
P-Persistent: Transmits with p probability when idle; used in Wifi
O-Persistent: Transmission based on pre-decided superiority/order
CSMA/CD (Collision Detection)
Adds collision handling to CSMA
Frame size must be large enough to detect collision while sending
Frame transmission delay >= 2x max propagation delay
Worst-case time wasted is 2Tp (contention slots)
CSMA/CA (Collision Avoidance)
Used where collision detection is difficult (e.g., wireless)
Avoidance Mechanisms
Interframe Space (IFS): Waits for IFS duration after medium becomes idle
Contention Window: Random slots chosen as wait time; doubles if busy
Acknowledgement: Sender re-transmits if ACK not received before timeout
Controlled Access Protocols
Only one device uses network at a time (taking turns)
Data sent only by station approved by others
Channelization
Bandwidth shared in time, frequency, and code
FDMA (Frequency Division Multiple Access)
Bandwidth divided into equal bands
Uses Guard bands to avoid crosstalk/overlap
TDMA (Time Division Multiple Access)
Time divided into slots allotted to stations
Synchronization overhead resolved by synchronization bits
Propagation delay resolved by guard bands
CDMA (Code Division Multiple Access)
One channel carries all transmissions simultaneously
Data transmitted simultaneously using different code languages
OFDMA (Orthogonal FDMA)
Bandwidth divided into small subcarriers
Used widely in 5G technology
Networking Devices and Concepts
Switch
Segments networks into subnets/LAN segments
Filters/forwards packets based on MAC address
Operates at Data Link Layer (OSI)
Features: Error checking, full duplex, limited bandwidth allocation
Supports Unicast, Multicast, Broadcast
Uses Packet-switching techniques
VLANs (Virtual LAN)
Logically divides devices on Layer 2
Divides broadcast domain using switches
Requires Inter Vlan routing to forward packets between VLANs
VLAN Ranges
0, 4095: Reserved
1: Default VLAN, cannot be deleted/edited
2-1001: Normal range (create, edit, delete allowed)
1002-1005: CISCO defaults (fddi/token rings), cannot be deleted
1006-4094: Extended range
VTP (VLAN Trunking Protocol)
CISCO proprietary protocol
Synchronizes VLAN information in the same VTP domain
Allows adding, deleting, renaming VLANs propagated to other switches
Requirements for VTP
Same VTP version
Same VTP domain name
One switch must be a server
Authentication must match
VTP Modes
Server: Default mode, allows CRUD operations, saved in NVRAM
Client: Receives/forwards updates, not saved in NVRAM
Transparent: Forwards summary advertisements, can make local database
STP (Spanning Tree Protocol)
IEEE standard 802.1D-1988
Generates a single spanning tree (loop-free active forwarding topology)
Exchanges Bridge Protocol Data Units (BPDUs)
Function: Computes loop-free topology (blocking redundant ports)
Function: Automatically recovers from switch failure by reconfiguring tree
RSTP (Rapid STP) is default mode on every port
Wide Area Network (WAN) Protocols/Technologies
Converged WAN
Integrates multiple traffic types (data, voice, video) over single infrastructure
Services Included
Internet access
VoIP
Video conferencing
Cloud services
Unified communication (UC)
Role of Wireless Routers
Provide seamless wireless connectivity
Enable Hybrid WANs (integrating 4G/5G with MPLS/Ethernet)
Serve as Backup and Failover connections
Perform Load Balancing over wired and wireless links
PPP (Point-to-Point Protocol)
Data link layer protocol (RFC 1661)
Transmits multiprotocol data between two directly connected computers
Byte-oriented, used in broadband communications
Components
Encapsulation Component (encapsulates datagram)
LCP (Link Control Protocol): Establishes, configures, terminates links
Authentication Protocols (AP)
PAP (Password Authentication Protocol)
CHAP (Challenge Handshake Authentication Protocol)
NCPs (Network Control Protocols): Negotiates parameters for network layer
IPCP
OSINLCP
IPXCP
IPV6CP
Frame Relay
Packet-switching network protocol (Data Link Layer)
Connects LANs and transmits data across WANs
Better alternative to point-to-point for multiple nodes
Allows transmission of different size packets
Uses virtual circuits set up by frame relay switches
Identifies destination via Data Link Connection Identifier (DLCI)
Lacks error control and flow management mechanism
ATM Networks (Asynchronous Transfer Mode)
ITU-T efficient for cell relay
Transmits fixed-size packets called cells (53 bytes: 5 header, 48 payload)
Connection-oriented, uses cell switching with virtual circuits
Can handle constant and variable rate traffic (QoS support)
Historical Perspective
Developed late 1980s/early 1990s
Uses Virtual Channels (VCs) and Virtual Paths (VPs) for connection
Switches use VPI/VCI for routing
Protocol Architecture
Virtual Path Connections (VPCs) bundle Virtual Channel Connections (VCCs)
ATM Adaption Layer (AAL): Prepares user data for cell conversion
Physical Layer: Manages medium-dependent transmission, converts cells to bitstream
ATM Layer: Handles switching, congestion control, cell multiplexing/relay (using VPI/VCI)
AAL Layer Protocols (Adaptation Layer)
Provides interface for non-ATM networks
Segments higher layer packets into 48-byte ATM cells
AAL Type 0: Simplest, raw cells (no guaranteed delivery)
AAL Type 1: Synchronous, constant rate traffic (voice)
AAL Type 2: Synchronous, variable rate traffic (wireless applications)
AAL Type 3/4: Variable rate data, connection-oriented/connectionless traffic
AAL Type 5 (SEAL): Simplified header, used for IP over ATM
Other Algorithms and Concepts
Bayesian Algorithm (Naïve Bayes)
Classification technique based on Bayes' Theorem
Assumes independence among predictors ('Naïve')
Simple to build, useful for large data sets
Bayes Theorem Equation Terms
P(c|x): Posterior chance (class given predictor)
P(c): Prior chance of class
P(x|c): Chance of predictor given class
P(x): Prior chance of predictor
FCFS Scheduling (First Come First Serve)
Simplest CPU scheduling algorithm
Process requesting CPU first is allocated first
Implemented using FIFO queue
Non-preemptive algorithm (also supports preemptive)
Easy to implement, but high wait time (inefficient performance)
Tree Topology (Networking)
Hierarchical arrangement
Central node known as the root
Root linked to multiple tiers of child nodes (branches)